

LAB 1

Lab: Identification and Use of Testing and Measuring Instruments in PC System and Peripherals Repair

1. Objectives

- Identify fundamental hardware testing and measuring instruments used in PC repair.
- Understand the core function and practical application of each instrument during component-level diagnostics.
- Learn standard safety protocols when working with electrical measurement equipment on PC hardware.

2. Safety Precautions

Electrical Safety Warning

Ensure the computer power supply is completely disconnected from the AC mains outlet before conducting continuity tests or handling internal motherboard components. Always wear an Anti-Static Wrist Strap connected to a grounded surface to prevent Electrostatic Discharge (ESD) damage.

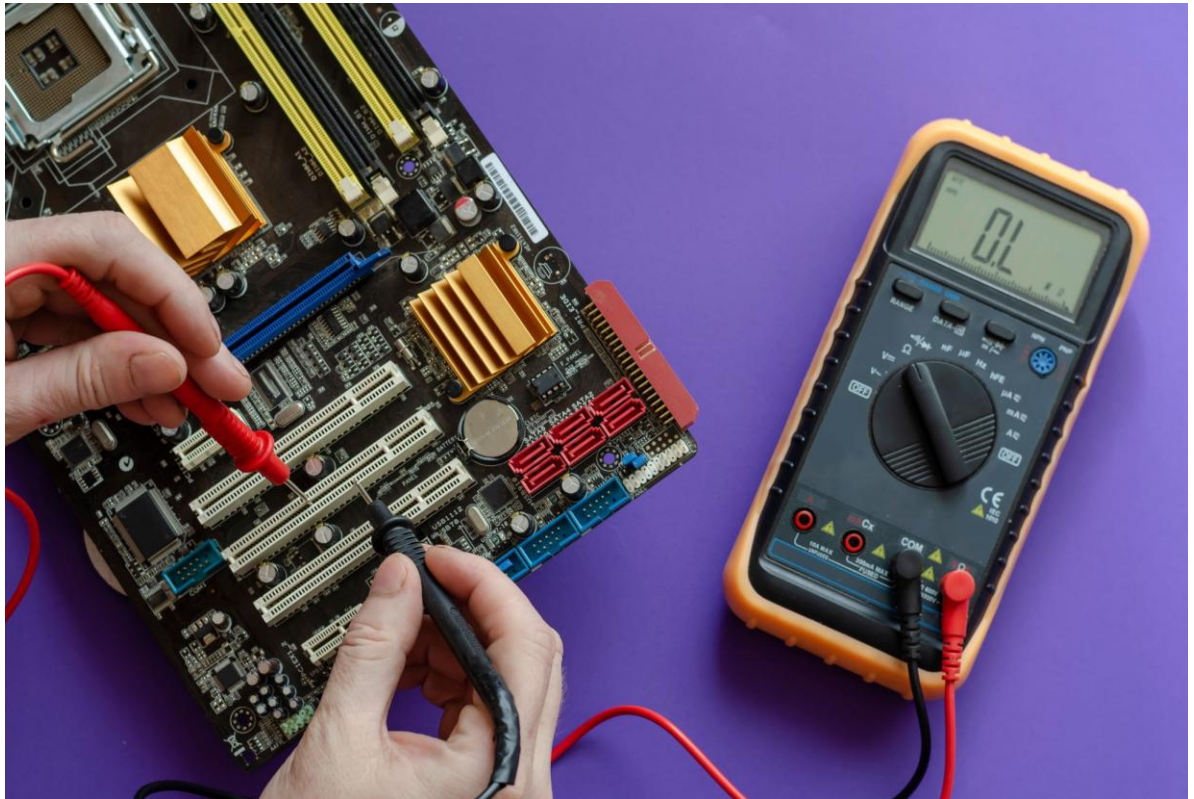
3. Equipment & Tools Required

1. Digital Multimeter (DMM) with test probes
2. Oscilloscope (Benchtop or Portable)
3. PCI / PCIe POST Diagnostic Card
4. Logic Probe
5. Anti-Static Wrist Strap and Mat
6. Disassembled PC Motherboard and ATX Power Supply Unit (PSU)

4. Theoretical Background & Key Instruments

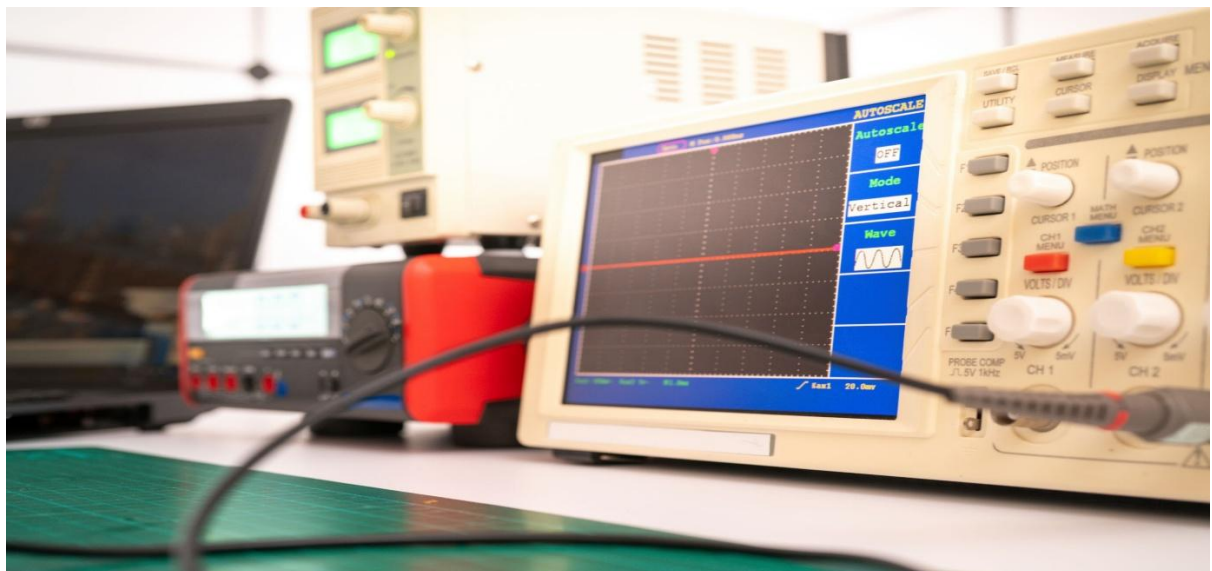
Instrument 1: Digital Multimeter (DMM)

- **Primary Function:** Measures DC/AC voltage, resistance, current, and circuit continuity.
- **Repair Application:** Used to test ATX Power Supply Unit (PSU) output voltage lines (+12V, +5V, +3.3V), check CMOS battery voltage level (3V nominal), locate short circuits using continuity mode, and verify discrete components (MOSFETs, resistors, diodes).



Instrument 2: Oscilloscope

- **Primary Function:** Measures and visualizes electrical signal waveforms over time.
- **Repair Application:** Analyses clock signals from system crystal oscillators, detects power ripple noise across Voltage Regulator Modules (VRMs), and monitors high-speed data line operations.



Instrument 3: POST (Power-On Self-Test) Diagnostic Card

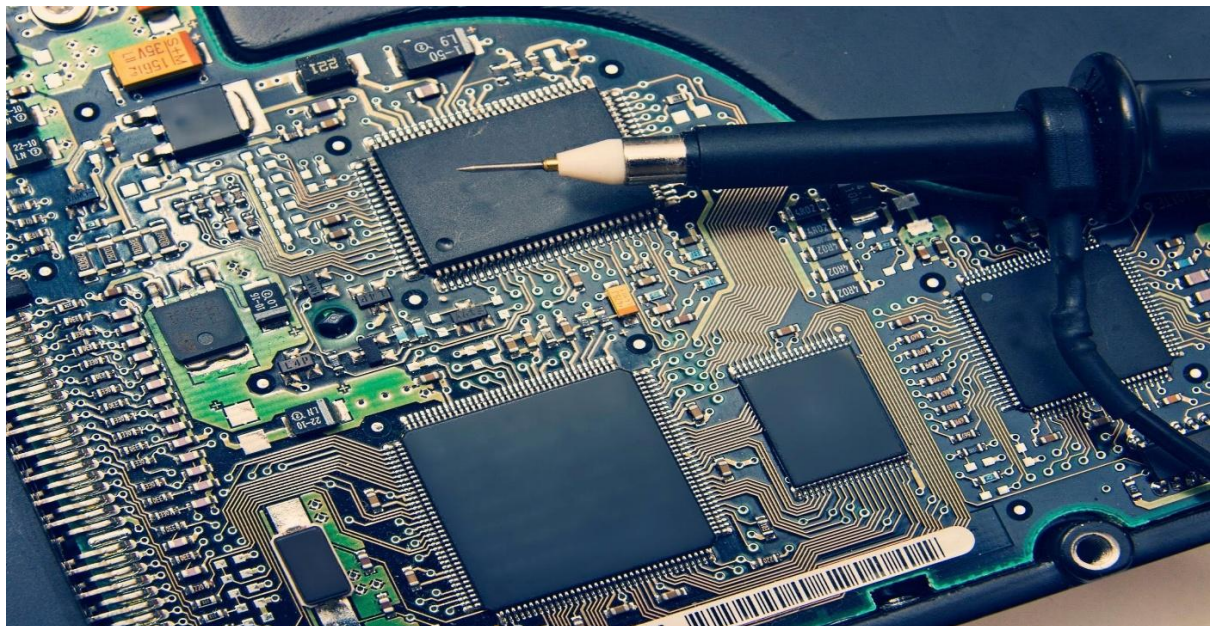
- **Primary Function:** Reads hexadecimal POST codes output by the system BIOS during startup sequences.

- **Repair Application:** Inserted into PCI, PCIe, or LPC expansion slots to isolate boot failures (e.g., RAM initialization failure, GPU detection fault, or CPU boot halt).



Instrument 4: Digital Logic Probe

- **Primary Function:** Indicates high, low, or pulsing logic states in digital circuits via LED indicators.
- **Repair Application:** Checks logic states on integrated circuit (IC) pins, system bus lines, and reset control lines without requiring a full waveform display.



5. Summary Reference Table

Instrument	Primary Metric Measured	Typical PC Fault Diagnosed
Digital Multimeter	Voltage (V), Resistance (Ω), Continuity	Dead PSU rails, shorted caps, low CMOS battery
Oscilloscope	Signal Waveform, Frequency (Hz), Noise	VRM voltage ripple, faulty clock oscillator
POST Card	Hexadecimal BIOS Debug Codes	System powers on (fans spin) but no video display
Logic Probe	Binary States (High / Low / Pulse)	Stuck IC lines, faulty reset switches, data bus activity

6. Lab Tasks & Procedure

1. Check CMOS Battery Voltage: Digital Multimeter.

Set the multimeter dial to DC Voltage (20V range). Place the red probe on the top (+ surface) of the CR2032 battery and the black probe on the outer side ground tab. Record the voltage reading.

2. Verify Motherboard Continuity: Continuity Mode.

Switch the DMM to Continuity mode (buzzer symbol). Touch probe tips together to confirm the audible tone. Place one probe on a ground standoff screw hole and the second probe on a GND pin of the 24-pin ATX connector to verify grounding integrity.

3. Read POST Codes: Diagnostic Card.

Insert the POST card into an available slot on a test bench motherboard. Power on the system and record the final 2-digit hexadecimal code displayed on the LED display. Refer to the motherboard BIOS manual to identify the corresponding boot state.

7. Observations & Results

1. CMOS Battery Measurement:

- Measured Voltage: _____ V
- Status (Pass $\geq 2.8\text{V}$ / Fail $< 2.8\text{V}$): _____

2. POST Diagnostic Result:

- Final Code Displayed: _____
- Indicated BIOS Stage / Fault: _____